

OMID ZANGANEH

GIS Developer & Spatial Data Analyst

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Professional Summary

GIS Developer and Spatial Data Analyst with 8+ years of experience in geospatial analysis, remote sensing, and custom GIS tool development. Expertise in Python, ArcGIS Pro SDK, and web-based mapping applications. Proven track record of delivering innovative solutions for telecom network design, environmental monitoring, and automated spatial analysis workflows.

Professional Experience

GIS Developer

May 2022 – Present

Rogers Communications Inc., London, ON

- Developed 25+ custom ArcGIS Pro tools using Python and C# .NET, automating complex geospatial workflows and reducing manual processing time by 80%
- Built an enterprise FTTH network designer tool that generates optimized fiber routes, calculates material requirements, and produces construction-ready maps, serving 150+ users
- Created automated bore profile generators that integrate civil engineering data with GIS, processing 500+ profiles monthly with 95% accuracy
- Designed and deployed web-based GIS tools using Next.js, React, and Leaflet for field data collection and real-time mapping
- Implemented AI-powered object detection pipelines using YOLO and OpenCV for automated feature extraction from aerial and street-level imagery
- Engineered ETL processes for enterprise spatial databases (SQL Server, PostgreSQL) handling millions of records with spatial indexes and optimized queries

GIS Analyst

September 2020 – May 2022

Rogers Communications Inc., London, ON

- Performed spatial analysis for telecom network expansion projects, identifying optimal service areas and infrastructure placement using advanced geoprocessing techniques
- Developed Python scripts for automated data validation and quality control, reducing errors by 70%
- Created interactive dashboards and maps for executive reporting using ArcGIS Online and Tableau
- Managed enterprise geodatabase schemas and implemented version control workflows for multi-user editing

Remote Sensing Specialist

January 2017 – August 2020

Iranian Space Agency, Tehran

- Processed and analyzed satellite imagery (Landsat, Sentinel-2, MODIS) for land cover classification, change detection, and environmental monitoring across 50,000+ km² study areas
- Developed automated image processing workflows using Google Earth Engine and Python, reducing processing time from weeks to hours
- Conducted spectral analysis and supervised classification for agricultural mapping, achieving 92% accuracy
- Collaborated with environmental scientists on climate impact studies, providing geospatial analysis & environmental justice.

Education

Master of Science, Remote Sensing and GIS

September 2019

K. N. Toosi University of Technology, Tehran

Thesis: Multi-temporal analysis of urban heat islands using thermal remote sensing & environmental justice.

Technical Skills

Languages & Frameworks: Python • C# / .NET • SQL • TypeScript • JavaScript • HTML/CSS • React • Next.js • WPF

GIS & Spatial: ArcGIS Pro • ArcGIS Pro SDK • ArcPy • Python Toolboxes (.pyt) • ArcGIS Online • ArcGIS Enterprise • QGIS • Geoprocessing • Network Analysis • Remote Sensing • Google Earth Engine

AI / ML: Azure OpenAI • Azure AI Foundry • Google AI Studio / Gemini • YOLO • OpenCV • Prompt Engineering • Web Grounding • Batch Classification Pipelines

Data & Backend: SQL Server • Supabase • PostgreSQL • pyodbc • ETL Pipelines • Spatial Types (GEOMETRY/GEOGRAPHY) • Smartsheet API

Web & Cloud: Next.js App Router • Vite • Leaflet • Vercel • REST APIs • PWA • Microsoft Azure • Google Cloud

Desktop & Tooling: CustomTkinter • PyInstaller • Playwright • Matplotlib • AutoCAD • ENVI • SNAP • Photomod • Tableau • Adobe Photoshop • Adobe Illustrator

Languages: English (Fluent) • Persian (Native)